Temperature & Top_P Explanation

The **temperature** parameter controls how deterministic or creative the model's responses are. A low temperature (0–0.3) makes the model more focused, factual, and consistent, it is ideal for technical or accuracy-sensitive tasks because the model chooses the most likely next words. A higher temperature (0.7–1.0) increases creativity and randomness, allowing the model to explore more varied responses but also increasing the risk of irrelevant or less accurate outputs. In this project, a **temperature of 0.2** was chosen to ensure that responses to heavy machinery technical questions remain stable, precise, and reliable.

The **top_p** parameter (nucleus sampling) narrows the model's word selection to only the smallest group of tokens whose combined probability reaches the top_p threshold. Lower values (e.g., 0.8) limit the model to highly probable words, making responses straightforward and controlled, while higher values (0.95–1.0) allow more diversity by including less likely tokens. For this project, **top_p was set to 0.9**, offering a balanced approach and ensuring technically accurate answers while still giving the model enough flexibility to generate natural, readable explanations.